Rui Gong

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology
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Research Interests

Semidefinite programming, combinatorial optimization, convex relaxations, stochastic and dynamic optimization, and computational methods for structured decision problems.

Education

Georgia Institute of Technology Atlanta, Georgia, U.S.A.
PhD in Operations Research 2023.08–2027.08 (expected)

- Advisor: Alejandro Toriello; co-advised by Diego Cifuentes.
- Passed comprehensive exams.

University of Waterloo Waterloo, Ontario, Canada
M.Math. in Combinatorics & Optimization 2021.09–2023.06

- Thesis: *Low-Rank Plus Sparse Decompositions of Large-Scale Matrices via Semidefinite Optimization.*
- Advisor: Levent Tuncel. Passed C&O PhD first-stage comprehensive exams in Continuous Optimization and Graph Theory.

University of Waterloo Waterloo, Ontario, Canada
B.Math. Triple Major in Mathematical Finance, Mathematical Optimization, Statistics 2017.09–2021.08

- Dean's Honour List.

Preprints and Manuscripts

Rounding the Lovász Theta Function with a Value Function Approximation
Rui Gong, Diego Cifuentes, Alejandro Toriello. Under major revision of SIAMOPT. arXiv:2504.07204

Dynamic Interval Scheduling with Random Start and End Times
Rui Gong, Alejandro Toriello. Under major revision of *Operations Research Letters*. arXiv:2602.07217

Work in Progress

Rounding the SDP Relaxations of the Quadratic Assignment Problem
Rui Gong, Mohit Singh, Alejandro Toriello. In progress.

Scholarships and Awards

- Ronald & Carol Beerman Fellowship, Georgia Institute of Technology, 2024–2025.
- ISyE Premium Fellowship, Georgia Institute of Technology.
- Ronald & Carol Beerman Fellowship, Georgia Institute of Technology, 2023–2024.
- Math Domestic Graduate Student Award, University of Waterloo, 2022–2023.
- Sinclair Graduate Scholarship, University of Waterloo, 2022–2023.
- C & O Graduate Award, University of Waterloo, 2022.01–2022.04.
- President's Scholarship, University of Waterloo, 2017.

Presentations

Conference Talks and Seminars

- *Dynamic Interval Scheduling with Random Start and End Times*, INFORMS Optimization Society Conference, Atlanta, U.S.A., March 20, 2026.
- *Rounding the Lovász Theta Function with a Value Function Approximation*, INFORMS Annual Meeting, Atlanta, U.S.A., October 26, 2025.
- *Value Function Approximation for Maximum Stable Set Problem*, Combinatorial Optimisation and Graph Theory Session, International Symposium on Mathematical Programming (ISMP 2024), Montréal, Canada, July 26, 2024.
- *Value Function Approximation for Maximum Stable Set Problem*, First Year PhD Seminar, ISyE, Georgia Institute of Technology, February 23, 2024.
- *Low-Rank Plus Sparse Decompositions of Large-Scale Matrices via Semidefinite Optimization*, M.Math. Thesis Talk, Department of Combinatorics & Optimization, University of Waterloo, May 3, 2023.

- *Convex Optimization, Factor Analysis and Realizability of a Subspace*, Graduate Research Seminar, Department of Combinatorics & Optimization, University of Waterloo, April 11, 2022.

Posters

- *Deriving Quadratic Assignment Solutions from Semidefinite Relaxations*, Mixed Integer Programming Workshop 2026, Poster Finalist, University of Connecticut, May 18, 2026.
- *Rounding the Lovász Theta Function with a Value Function Approximation*, Mixed Integer Programming Workshop 2024, Poster Finalist, University of Kentucky, March 4, 2024.

Workshops Attended: Fulkerson 100, University of Waterloo, July 17–19, 2024; Algorithms, Combinatorics and Optimization Research Network (ACORN) Meeting 2023, Georgia Institute of Technology, March 9–11, 2023; 24th Midwest Optimization Meeting & Workshop on Large Scale Optimization and Applications, University of Waterloo, October 28–29, 2022.

Teaching

Teaching Assistant, University of Waterloo

- Portfolio Optimization (CO372), undergraduate course, 2021.09–2022.04 and 2022.09–2022.12. Held office hours and marked assignments and exams.
- Semidefinite Programming (CO471/671), graduate course, 2022.05–2022.08. Marked assignments.

Industry Experience

Intact Financial Corporation Toronto, Canada

Actuarial Analyst, Ontario PL Pricing 2020.01–2020.04

Queried and processed insurance pricing datasets in SAS to support internal actuarial analyses; produced recurring earnings and loss reports in Excel/SAS; validated rating-algorithm outputs using R.

Technical Skills

Programming and tools: Julia, Python; LaTeX.
Optimization and computing: JuMP, Gurobi, COPT, Mosek, COSMO, SCS, HiGHS; Slurm/HPC workflows.
Languages: Mandarin Chinese native; English fluent.